Technical Report

TPT Data Explorers 2026

Data Explorers Team

May 30, 2026

Abstract

The TPT Data Explorers 2026 project addresses crucial supply chain management and dealer behavior analysis challenges for Thong Nhat Bike. Amidst market volatility and a severe 9-month data gap spanning from April 2025 to December 2025, this technical report presents three core analytical tracks intended to stabilize the business forecasting baseline. First, the Product Demand Forecasting track for Q2 2026 navigates the March 2026 Demand Shock to optimize procurement. Second, an in-depth trend analysis of product colors and variants provides actionable inventory insights. Third, the prediction of dealer ordering behavior leverages RFM B2B segmentation coupled with classification modeling. A hybrid approach utilizing Group-Share Proportional methodologies alongside experimental Machine Learning sets provides a reproducible operational baseline, while strictly acknowledging systemic constraints due to historical data sparsity.

Keywords: Supply Chain, Demand Forecasting, RFM B2B, Machine Learning, Inventory Optimization.

Contents

1 Introduction

Thong Nhat Bike faces an immediate operational need to systematically forecast the ordering behavior of its diverse B2B dealer network. Accurate forecasting not only optimizes working capital tied up in inventory but also directs targeted customer care campaigns efficiently. A successful forecasting system must delicately balance algorithmic sensitivity with clear business interpretability.

The project structurally focuses on three primary tracks. Track 1 forecasts volume allocation by SKU to preempt supply chain disruptions. Track 2 analyzes underlying color trends, acting as a critical attribute affecting stock turnover rates. Track 3 systematically scores and ranks dealer priorities to support customer care resource allocation, leveraging behavioral RFM modeling to personalize engagement strategies.

2 System Architecture

The system is designed with a strict modular approach, ensuring individual components can evolve independently. The Database Layer utilizes PostgreSQL as the central reliable storage mechanism. Ingestion processes handle both historical data seeds and automated pipelines that extract unstructured transactions from emails and PDFs.

Figure 1 illustrates the holistic Data Flow. Extracted data undergoes a robust ETL layer to patch structural and historical inconsistencies. The Feature Store systematically transforms transactional records into continuous Time-series Panels. All forecasting artifacts are eventually exported as CSV datasets and accessible SQL Views.

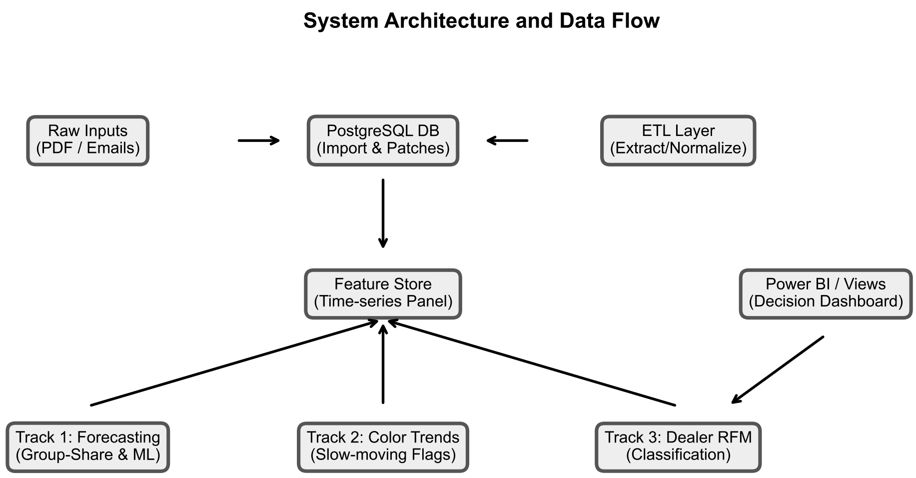

Figure 1: System Architecture and Data Flow Diagram

The clear separation of concerns inherent in this layered architecture supports maintainability and codebase reusability for future cyclical updates.

3 Database and ETL Pipeline

The analytical foundation begins with establishing a robust database and ETL (Extract, Transform, Load) pipeline, detailed in Table 1. The database and ETL layer represents the foundation of the project: schema creation, historical import, data cleaning patches, email/PDF extraction, normalization, and BI-serving SQL views are upstream dependencies for the forecasting pipeline.

Table 1: Pipeline Components and ETL Scripts

Component	**Script Alias**	**Role**
Database Init	create-tables.sql	Initializes PostgreSQL schema
Historical Seed	import-data.sql	Ingests historical data
ETL Parser	extract-validate.py	Automated PDF/Email parsing
Data Patches	sql-patches.sql	Fixes geo and color errors
Feature Store	build-features.py	Constructs Time-series
Track 1 Model	core-baselines.py	Group-Share Forecast
Track 2 Model	color-trends.py	Color trend allocation
Track 3 Model	rfm-track3.py	RFM B2B ranking
BI Dashboard	v-rfm-analysis.sql	Direct views for Power BI

Manual Data Quality Patches (SQL) addressed known color and geographic inconsistencies. For example, harmonizing misspelled color variants prevented the artificial splintering of a single product's demand signal into multiple disjointed time-series.

4 Data Quality Audit

A Data Quality Audit clarified the overarching data landscape, exposing both structural strengths and historical flaws. Table 2 itemizes critical metrics shaping the system's design constraints.

Table 2: Key Data Facts and Implications

Metric	Value	Characteristic	Implication
Order lines	25,754	Transactional	Revenue calculation base
Sales Orders	2,759	Transactional	Frequency evaluation
Dealers	798	Master Data	Primary RFM B2B input
SKUs (Products)	265	Master Data	Target universe
Total Revenue	109,445,161,439 VND	Transactional	Historical revenue baseline
Total Quantity	72,146	Transactional	Historical volume
Mar 2026 Orders	1,132	Demand Shock	Creates March Shock
Data Gap	04/25 - 12/25	Limitation	Needs block-aware lag
Unknown Hierarchy	55	Metadata Error	Tagged UNKNOWN

4.1 Data Limitations and Constraints

The most prominent anomaly is a substantial temporal gap spanning exactly 9 months (April 2025 to December 2025). This gap prevents standard YoY comparisons and invalidates classical cyclical forecasting models like SARIMA. Instead of fabricating data through arbitrary interpolation, the project accepts this discontinuity and enforces block-aware lagging strategies to isolate 2025 data.

4.2 UNKNOWN Hierarchy Impact

The audit additionally identified 55 SKUs lacking proper hierarchy assignment; these were tagged as UNKNOWN to reduce downstream hierarchy distortion. The implications of this missing metadata are significant across multiple business streams, as outlined in Table 3.

Table 3: Impact of UNKNOWN Hierarchy on Downstream Analytics

Impact Area	Description
Group-Share Allocation	These 55 SKUs cannot be assigned to any product group, reducing group-share precision
Color Trend Analysis	Missing hierarchy prevents proper color-category cross-tabulation
Feature Engineering	line_id_clean and group_code_encoded default to UNKNOWN, weakening ML signal

Until the master data is corrected, these products are segregated into a pseudo-group to ensure they do not contaminate the primary group-share allocation metrics.

5 Feature Engineering and Alignment

To prevent Data Leakage—an algorithmic flaw where models are implicitly fed future information—the feature store strictly aligns historical training rows with future cutoff constraints. No target variable data from April 2026 onwards leaks into the prediction horizon. Furthermore, any variables reflecting current-period pricing are strictly prohibited to reduce forward-looking bias. During feature engineering, `is_zero_sales_row` is audit-only and excluded from model candidates. Future Q2 rows use cutoff-known features only. Train/future feature alignment is fail-fast to avoid silent feature dropping. Table 4 outlines the primary guardrails enforcing this separation.

Table 4: Feature Engineering Guardrails

Feature Family	**Alias**	**Rule**	**Reason**
Zero-sales Panel	is-zero-sales	Mandatory	Ensures continuous time-series
Gap-crossing Lags	qty-lag-1	Restricted	Prevents signal leakage
Current Price	current-price	Banned	Causes forward-looking bias
Schema Alignment	is-aligned	Mandatory	Schema must match exactly

6 Forecasting Strategy

Due to the severely truncated historical timeframe and the 9-month data blackout in 2025, cyclical models such as ARIMA or Prophet are not statistically reliable under the current data constraints. Deploying them would invite significant forecast risk. The primary business forecasting methodology deployed is Group-Share Proportional allocation.

In parallel, Machine Learning tree-based models (LightGBM/CatBoost) are deployed as an Enhanced Baseline experiment to capture non-linear relationships. However, due to the unprecedented March 2026 Demand Shock, utilizing March as a validation set introduces substantial noise, causing the models to potentially overreact. Consequently, Group-Share remains the primary operational forecast, while ML results serve exclusively as volatility reference material.

6.1 Modeling Validation

During the Machine Learning evaluation phase, variants including LightGBM, XGBoost, and CatBoost were evaluated under a constrained temporal holdout / stress-validation setup, with March 2026 treated as the stress validation period. Table 5 compares these models across the overall segment using WMAPE (Weighted Mean Absolute Percentage Error).

Table 5: Model Comparison Metrics (Overall)

Model Variant	MAE	RMSE	WMAPE	Bias Ratio
LightGBM_monthly_minimal_features	61.35	116.83	0.4744	-0.1653
XGBoost_monthly_minimal_features	64.54	107.33	0.4990	0.0178
CatBoost_monthly_minimal_features	55.48	97.88	**0.4290**	-0.0036
LightGBM_monthly_extended_features	67.44	124.95	0.5215	-0.3801
XGBoost_monthly_extended_features	62.28	105.84	0.4816	-0.2373
CatBoost_monthly_extended_features	59.27	108.35	0.4583	-0.2863
LightGBM_weekly_minimal_features	31.18	49.16	0.9460	0.3621
XGBoost_weekly_minimal_features	29.41	48.62	0.8923	0.2648
CatBoost_weekly_minimal_features	31.08	52.62	0.9431	0.2770
LightGBM_weekly_extended_features	29.28	47.74	0.8884	0.2409
XGBoost_weekly_extended_features	29.17	48.12	0.8850	0.2616
CatBoost_weekly_extended_features	29.33	49.38	0.8901	0.2530

Note. WMAPE is the primary normalized metric for comparison.

Table 6: Best ML Forecast vs Group-Share Base

Method	Q2 Forecast Quantity	Q2 Forecast Revenue (VND)	Interpretation
Group-Share Base	37,596	60,914,780,907	Primary operational baseline (Base scenario)
CatBoost_monthly_minimal_features	51,640	85,652,196,872	Best ML validation variant (reference only)

Note. Group-Share Base is taken from the Base scenario only, not the sum of Conservative, Base, and Aggressive scenarios. CatBoost is shown as a reference ML forecast and is not the official planning baseline.

CatBoost_monthly_minimal_features achieved the lowest validation WMAPE among the evaluated ML variants; however, Group-Share remains the primary operational forecast.

It is important to interpret these metrics with caution. While weekly models utilize a larger volume of data points, their WMAPE is often worse than monthly models because demand is highly volatile at the weekly grain, leading to higher variance in error margins. Furthermore, absolute metrics like MAE and RMSE are not directly comparable between weekly and monthly models due to the differing magnitudes of aggregated volumes.

6.2 Machine Learning Insights

Despite the aforementioned volatility, the Machine Learning experiments extracted valuable feature importance signals. Table 7 highlights the core features driving the ML algorithms.

Table 7: Top ML Features by Importance

Feature	Avg Importance	Interpretation
qty_roll_mean_4w	92.7124	Rolling weekly demand momentum
qty_lag_2w	72.8203	Two-week lagged demand signal
qty_lag_1	65.3358	Immediate short-term demand momentum
qty_lag_4w	65.2720	Four-week lagged demand signal
line_id_clean	56.1627	Product hierarchy / category proxy

Note. Feature importance is summarized for models with comparable importance outputs and is aggregated across monthly and weekly ML variants. It is directional and should not be interpreted causally.

6.3 Business Scenario Planning

Due to ML volatility caused by the March shock, Group-Share remains the primary operational forecast. Table 8 and Figure 2 summarize the Group-Share allocation scenarios for Q2 2026.

Table 8: Scenario Delta and Planning Implication

Scenario	Q2 Quantity	Q2 Revenue (VND)	Delta Qty vs Base	Delta Revenue vs Base (VND)	Planning Use
Conservative	31,956	51,776,591,623 VND	-5,640	-9,138,189,284 VND	Financial risk control
Base	37,596	60,914,780,907 VND	0	0 VND	Primary baseline
Aggressive	42,953	69,594,440,480 VND	+5,357	+8,679,659,573 VND	Capacity buffer

Note. Conservative controls financial downside risk. Aggressive prepares for sales upside.

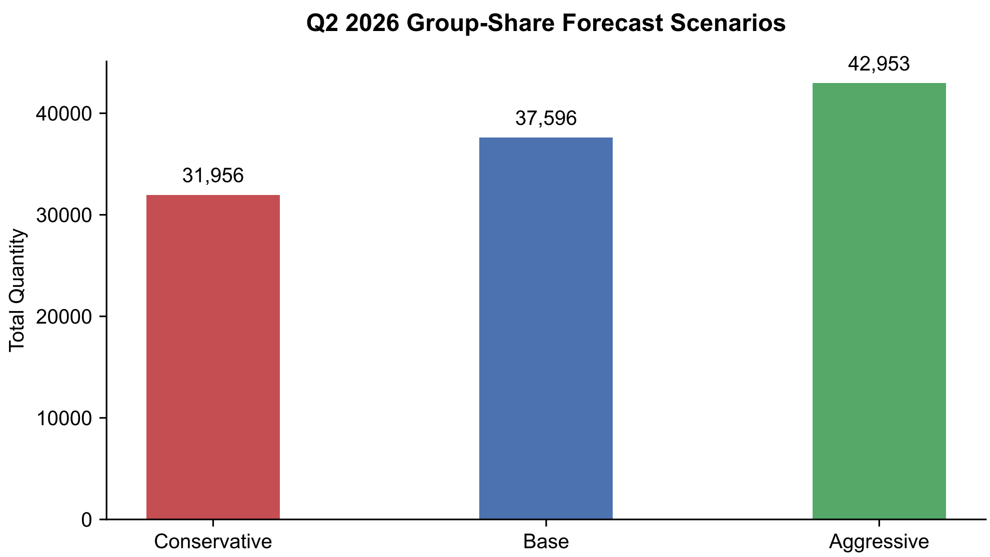

Figure 2: Q2 2026 Group-Share Forecasted Quantity Scenarios

The Base scenario anchors standard material procurement planning. The Aggressive scenario serves as a strategic capacity buffer for supply chain resilience. Finally, the Conservative scenario assists executive management in controlling cash flow and inventory risks.

7 Track 2: Color Analysis

Within the bicycle distribution industry, the base color attribute acts as a useful dealer order signal reflecting localized predictions of end-consumer preferences by dealers. Table 9 and Figure 3 illustrate the Top 10 primary colors dominating the upcoming procurement cycle.

7.1 Color Contribution and Inventory Implications

Top colors listed below serve as strong candidates for procurement assurance, heavily driving the bulk of inventory planning. Conversely, low or declining colors require an inventory watch to mitigate stagnant stock. It is crucial to note that this color forecast exclusively reflects a dealer order signal, and should not be perfectly conflated with absolute end-consumer preference.

Table 9: Top Color Contribution Summary

Color	Forecast Qty	Qty Share	Forecast Rev (VND)	Rev Share
Cream	6,549	17.4%	10,178,738,540	16.7%
Black	5,797	15.4%	10,170,423,274	16.7%
Gray	4,308	11.5%	8,112,828,292	13.3%
Pink	3,816	10.1%	5,486,401,943	9.0%
Green	3,253	8.7%	5,418,630,863	8.9%
Blue	3,136	8.3%	4,786,143,217	7.9%
White	2,857	7.6%	4,997,821,606	8.2%
Brown	2,018	5.4%	3,055,462,552	5.0%
Mint	1,968	5.2%	3,005,926,217	4.9%
Red	1,707	4.5%	2,520,059,229	4.1%

Note. Source: outputs/modeling/phase3_color_summary_q2_2026.csv (columns: Group_Share_Base_Qty, Group_Share_Base_Rev). Forecast revenue is computed from the Group-Share proportional allocation applied to each base color. Qty Share and Rev Share are computed as a percentage of the total Q2 Base scenario forecast (37,596 units / 60,914,780,907 VND).

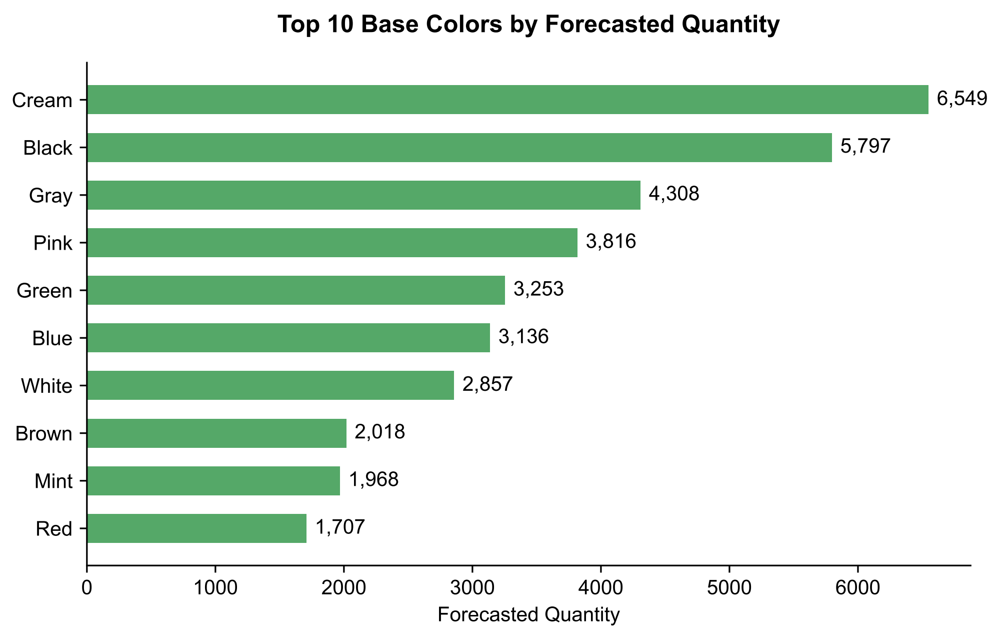

Figure 3: Top 10 Base Colors by Forecasted Quantity (Base Scenario)

Colors falling outside this Top 10 tier, or those exhibiting low or declining historical demand, are systematically flagged as slow-moving SKUs. This automated flagging empowers the sales department to execute early inventory liquidations, phasing out unviable variants before they negatively impact warehouse capacity.

8 Track 3: Dealer Ranking

Based on the established RFM B2B (Recency, Frequency, Monetary) paradigm, the system categorized the entire network of 798 dealers into 8 strategic segments. These segments were combined with a risk classification model to establish a unified Priority Ranking.

8.1 Dealer Segment Value Analysis

To understand the financial implications of these segments, Table 10 and Figure 4 describe the monetary distribution across the network. Revenue by dealer segment is directly calculated using available monetary metrics from the RFM processing phase.

Table 10: Dealer Segment Value Summary

Segment	Dealer Count	Historical Revenue (VND)	Revenue Share	Avg Revenue per Dealer (VND)	Avg Priority
Champions	57	46,249,985,880	42.3%	811,403,261	65.44
Loyal	132	25,620,749,540	23.4%	194,096,587	53.29
Potential	154	12,990,200,917	11.9%	84,351,954	31.77
Hibernating	158	9,233,353,836	8.4%	58,438,948	10.63
At Risk	25	6,743,037,357	6.2%	269,721,494	28.81
New	110	3,200,997,308	2.9%	29,099,976	9.59
Lost	159	3,030,809,439	2.8%	19,061,695	2.73
Big Spender	3	2,376,027,162	2.2%	792,009,054	27.58

Note. Source: outputs/modeling/phase3_dealer_priority_ranking_q2_2026.csv (columns: rfm_segment, monetary, marketing_priority_score). Segment monetary totals reconcile to the historical revenue total of 109,445,161,439 VND. Avg Priority is an internal ranking score and should not be interpreted as a calibrated probability. The segment distribution is based on the final dealer priority ranking artifact. It may differ from the dashboard RFM view if different cutoffs or scoring rules are used.

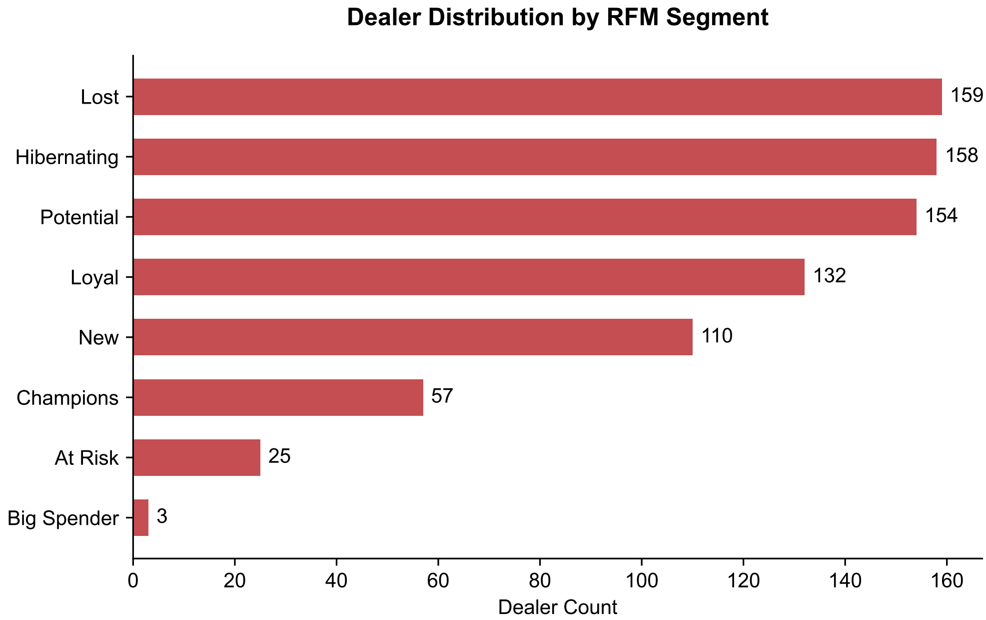

Figure 4: Dealer Distribution by RFM Segment

The Champions and Loyal segments act as the primary, stable revenue drivers. The Big Spender demographic requires a dedicated Key Account Care approach, as their purchasing patterns are voluminous but infrequent. Forcing aggressive sales frequency upon this group could prove counterproductive. Conversely, the high volume of Lost and At Risk accounts necessitates immediate reactivation campaigns.

8.2 Strategic Action Matrix

To translate these insights into actionable business strategies, Table 11 outlines the recommended approaches and Key Performance Indicators (KPIs) to monitor for each primary segment.

Table 11: Dealer Action Strategy Matrix

Segment	Business Meaning	Risk	Recommended Action	KPI to Monitor
Champions	High value, frequent	Low	Retain and protect	Retention rate / Repeat revenue
Loyal	Steady buyers	Low	Maintain engagement	Order frequency
Big Spender	High value, rare	Medium	Dedicated account care	Total revenue / Account retention
Potential	Promising buyers	Medium	Nurture and convert	Conversion to Loyal/Champion
At Risk	Dropping frequency	High	Reactivation campaign	Reactivation rate
Lost	Inactive long term	Very High	Low-cost reactivation or deprioritize	Response rate
Hibernating	Inactive	High	Periodic nurture	Reactivation signal
New	Just onboarded	Unknown	Onboarding	Second order rate

Note. Strategic actions and KPIs tailored to each RFM B2B segment.

9 Dashboard and Deliverables

All analytical pipelines converge into final output Artifacts designed for immediate business consumption. Table 12 outlines which artifacts are designated for various stakeholder groups.

Table 12: Which Artifact Should Each Stakeholder Use?

Stakeholder	Primary Need	Recommended Artifacts
Executive Team	Strategic overview	Executive Summary, Scenario Delta Table
Supply Chain / Procurement	Volume planning	Group-Share Forecast, Color Forecast
Sales / Trade Marketing	Dealer prioritization	Dealer Priority Ranking, Action Matrix
Data Science	Model stability	Feature Importance, ML Model Comparison

Note. Maps final outputs to specific operational workflows.

For BI visualization, the system architecture connects Power BI directly into the data warehouse via optimized views such as `v-rfm-analysis`. This ensures dashboards reflect real-time analytical updates.

10 Pipeline Reproducibility

Pipeline reproducibility ensures that historical validations can be re-run consistently. The Python architecture operates purely via CLI arguments to enforce immutable rules. Raw CSVs, interim data, feature stores, cache files, and model binaries are intentionally excluded from version control; the pipeline runner regenerates derived artifacts locally.

```
# 1. Dry-run to validate pipeline structure without overwriting
python src/models/forecasting/run_end_to_end.py --dry-run

# 2. End-to-End execution
python src/models/forecasting/run_end_to_end.py --allow-modeling --allow-
    overwrite
```

11 Limitations and Future Work

While providing a stable operational baseline, the system carries fundamental technical limitations that dictate clear avenues for future enhancement:

- **UNKNOWN Product Hierarchy**: 55 SKUs remain unmapped to known product groups or lines. Future work should prioritize master data correction to improve group-level allocation, BI drilldown, and model category signals.
- **Short History**: The absence of 9 core months of data in 2025 severely restricts cyclical analysis, requiring additional real historical data or carefully documented scenario assumptions in future iterations.
- **March Shock**: The sudden demand surge may incite overfitting phenomena in short-term forecasting if tree weights are not heavily regularized.
- **Cold-start Problem**: Newly onboarded dealers lack sufficient transactional history, inhibiting the generation of meaningful RFM behavioral scores.
- **Probability Calibration**: The Track 3 Classifier currently emits uncalibrated probabilities. Integrating Isotonic Regression or Platt Scaling is necessary to improve churn risk reliability.